

Worked for 4s

Absolutely. I've updated the specification to **Draft v0.2**, keeping your original structure and requirements, while adding the technical details that were missing for a developer to implement the system properly.

The main additions are **business rules, transaction status models, acceptance criteria, data constraints, organizational data scope, API standards, document controls, audit requirements, testing, environments, observability, backup/DR, and clearer offline boundaries**. These additions directly address the gaps identified in the system review.

LisheBora_System_Document_Revie...

LisheBora e-Sourcing Platform – System Requirements and Architecture Speci...

LISHEBORA e-SOURCING PLATFORM

SYSTEM REQUIREMENTS AND ARCHITECTURE SPECIFICATION

STEP School Feeding Project

Draft Version 0.2

Date: 23 September 2026

1. PURPOSE AND SCOPE

1.1 Purpose

The LisheBora e-Sourcing Platform is a digital school food procurement and supply-chain management platform designed to connect school food demand with qualified local suppliers.

The platform will support the full operational cycle from school meal planning and demand generation through budgeting, sourcing, supplier evaluation, contracting, aggregation, quality assurance, inventory, logistics, delivery, invoicing, payment tracking and performance monitoring.

The platform is intended to improve:

- Visibility and predictability of school food demand.
- Access to procurement opportunities for local suppliers.
- Efficiency and transparency of procurement processes.

- Food quality, safety and traceability.
- Delivery and stock management.
- Supplier payment tracking.
- Accountability through digital workflows and audit trails.
- Evidence generation for programme management and MEAL.
- Inclusive procurement participation by women, youth, persons with disabilities and other target producer groups.

The platform will complement existing government procurement, financial, school management and payment systems rather than replace statutory systems.

The source review recommends treating the existing brief as a strong business and functional foundation and expanding it into a detailed technical specification before full-scale development.

1.2 Scope

The platform will cover the following functional domains:

1. Identity and Access Management
2. Master Data Management
3. Demand and Nutrition Management
4. Budget Management
5. Supplier Management
6. Procurement and e-Sourcing
7. Contract Management
8. Aggregation and Quality Assurance
9. Inventory and Warehousing
10. Logistics and Delivery
11. Finance
12. Complaints and Performance Management
13. MEAL and Analytics
14. Platform Services

Cross-cutting capabilities will include:

- Risk and exception management
- Notifications
- Document management
- Audit logging
- Offline operation

- Data import/export
- API integration
- Security and privacy
- Reporting
- Monitoring and observability

Advanced capabilities such as GIS, QR/barcode traceability, advanced forecasting, AI-assisted procurement and advanced analytics will be architecturally supported but may be introduced in later phases.

The 14-domain structure follows the system review recommendation.

1.3 Out of Scope

The platform will not:

- Replace statutory government procurement systems.
- Replace national or county financial management systems.
- Replace national payment gateways.
- Replace school academic information systems.
- Act as a banking platform.
- Create independent procurement rules where statutory rules already exist.

Where such systems are required, LisheBora will integrate with them through APIs, secure file exchange or other agreed mechanisms.

2. FUNCTIONAL REQUIREMENTS

2.1 Identity and Access Management

FR-IAM-01 User Registration and Authentication

The system shall support:

- User registration.
- Username/email/mobile-based authentication.
- Password management.
- Password reset.
- Account activation and deactivation.
- Session management.

- Optional single sign-on through OAuth2/OIDC.

FR-IAM-02 Role-Based Access Control

The system shall support granular permissions including:

- Create
- View
- Edit
- Submit
- Verify
- Evaluate
- Approve
- Reject
- Pay
- Export
- Audit

Permissions shall be assigned based on both:

Role + Organizational/Data Scope

Example:

A County Procurement Officer may evaluate procurement activities within an assigned county but should not automatically access procurement records belonging to another county.

FR-IAM-03 Multi-Factor Authentication

MFA shall be configurable for:

- System administrators.
- Finance users.
- Approval officers.
- Procurement users.
- Other privileged users.

Final mandatory MFA roles shall be confirmed during inception.

FR-IAM-04 Session and Account Security

The system shall support:

- Configurable session timeout.
- Failed-login protection.
- Account lockout.
- Password policy.
- Session revocation.
- Device/session visibility where appropriate.

FR-IAM-05 Role Assignment

Authorized administrators shall be able to:

- Assign roles.
- Suspend roles.
- Revoke roles.
- Delegate authority.
- Configure acting officers.
- View role history.

FR-IAM-06 Segregation of Duties

The system shall prevent prohibited combinations of activities where configured.

Example:

A user who creates a purchase order should not approve the same purchase order where segregation-of-duties rules prohibit this.

2.2 MASTER DATA MANAGEMENT

FR-MD-01 Geographic and Organizational Structure

The system shall maintain:

- Counties.
- Sub-counties.
- Wards where required.
- School clusters.
- Schools.
- Aggregation centres.
- Warehouses.
- Suppliers.

- Partner organizations.

Each organizational entity shall have a unique identifier.

FR-MD-02 Commodity Catalogue

The system shall maintain:

- Commodity name.
- Commodity category.
- Unit of measure.
- Standard package size.
- Nutrition information.
- Quality specifications.
- Approved supplier requirements.
- Active/inactive status.

FR-MD-03 Nutrition and Menu Standards

The system shall support:

- Menu templates.
- Portion sizes.
- Feeding days.
- Nutrition specifications.
- Commodity requirements.
- Approved substitutions where applicable.

FR-MD-04 Configurable Parameters

Authorized administrators shall configure:

- Procurement thresholds.
- Approval thresholds.
- Budget rules.
- Quality standards.
- Supplier eligibility rules.
- Workflow rules.
- Notification rules.
- SLA periods.

FR-MD-05 Master Data Governance

All critical master data shall support:

- Versioning.
 - Effective dates.
 - Change history.
 - Approval where required.
 - Audit trail.
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2.3 DEMAND AND NUTRITION MANAGEMENT

FR-DEM-01 School Demand Entry

Schools shall capture:

- Enrolment.
- Expected attendance.
- Feeding days.
- Menu.
- Opening stock.
- Closing stock where required.
- Commodity requirements.

FR-DEM-02 Demand Calculation

The system shall calculate commodity requirements based on configured:

- Number of learners.
- Feeding days.
- Portion sizes.
- Menu.
- Existing stock.
- Approved wastage factors where applicable.

FR-DEM-03 Demand Validation

The system shall identify:

- Missing information.
- Unusual quantities.
- Negative stock.
- Excessive requirements.
- Missing menu information.
- Commodity/menu inconsistencies.

FR-DEM-04 Demand Consolidation

Demand shall be consolidated by configurable levels such as:

- School.
- Cluster.
- Sub-county.
- County.
- Procurement lot.

FR-DEM-05 Nutrition Validation

The system shall flag cases where configured nutrition requirements are not met.

FR-DEM-06 Procurement Plan

Approved demand shall be converted into a procurement plan containing:

- Commodity.
 - Quantity.
 - Unit.
 - Required delivery period.
 - School/cluster.
 - Estimated value.
 - Funding source.
 - Procurement method.
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2.4 BUDGET MANAGEMENT

FR-BUD-01 Budget Setup

The system shall support budgets by:

- Funding source.
- County.
- School.
- Programme.
- Commodity/category.
- Financial period.

FR-BUD-02 Budget Status

The system shall display:

- Approved budget.
- Committed amount.
- Obligated amount where applicable.
- Invoiced amount.
- Paid amount.
- Available balance.

The exact financial terminology and treatment of commitment versus expenditure shall be confirmed with the Finance team.

FR-BUD-03 Budget Validation

The system shall check available budget before:

- Procurement initiation where required.
- Purchase order creation.
- Contract/call-off commitment.

FR-BUD-04 Commitment Control

The system shall create budget commitments against approved transactions.

FR-BUD-05 Budget Alerts

The system shall generate alerts for:

- Low available balance.
- Approaching budget limit.
- Over-commitment.
- Expired budget.
- Unusual utilization.

FR-BUD-06 Exception Approval

Authorized users shall be able to process budget exceptions where permitted.

FR-BUD-07 Budget Reporting

Reports shall include:

- Budget utilization.
- Commitments.
- Available balances.
- Variances.
- Exceptions.

2.5 SUPPLIER 360° MANAGEMENT

FR-SUP-01 Supplier Registration

Suppliers shall register directly or through assisted registration.

Supplier information shall include:

- Legal/business name.
- Contact details.
- Ownership.
- Location.
- Supplier type.
- Bank/payment information where permitted.
- Inclusion attributes.
- Commodity categories.
- Capacity.

FR-SUP-02 Supplier Documents

The system shall maintain:

- Registration documents.
- Certificates.
- Licences.
- Tax documents.
- Compliance documents.
- Bank documents where required.

Documents shall support expiry dates.

FR-SUP-03 Supplier Prequalification

The system shall support:

- Application.
- Verification.
- Approval.
- Rejection.
- Suspension.
- Expiry.
- Renewal.

FR-SUP-04 Supplier Eligibility

Only suppliers meeting configured eligibility requirements shall participate in restricted procurement processes.

FR-SUP-05 Supplier 360° Profile

The supplier profile shall provide authorized users with a consolidated view of:

- Registration.
- Documents.
- Prequalification.
- Bids.
- Awards.
- Contracts.
- Deliveries.
- Quality history.
- Complaints.
- Invoices.
- Payments.
- Performance.

This supplier 360° capability responds directly to the system review recommendation.

FR-SUP-06 Supplier Performance Scorecard

The system shall calculate configurable indicators including:

- On-time delivery.
- Quantity fulfilment.
- Quality acceptance.
- Rejection rate.
- Complaint rate.
- Invoice/payment history.

FR-SUP-07 Supplier Status

Supplier status shall include configurable states such as:

- Draft.
- Submitted.
- Under Review.
- Approved.

- Prequalified.
 - Active.
 - Suspended.
 - Expired.
 - Rejected.
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2.6 PROCUREMENT AND e-SOURCING

FR-PRO-01 Procurement Planning

The system shall create procurement plans from approved demand.

FR-PRO-02 Sourcing Events

Authorized users shall create:

- RFQs.
- Competitive sourcing events.
- Lots.
- Framework sourcing.
- Call-offs.
- Other approved procurement methods.

FR-PRO-03 Supplier Notification

Eligible suppliers shall receive sourcing notifications through configured channels.

FR-PRO-04 Bid Submission

Suppliers shall submit:

- Price.
- Quantity.
- Delivery information.
- Required documents.
- Technical information where applicable.

FR-PRO-05 Bid Lock and Closing

At the configured closing time:

- Bid submission shall close automatically.
- Late submissions shall be blocked unless an authorized exception is recorded.

- Submitted bids shall be protected from unauthorized modification.

FR-PRO-06 Evaluation

The system shall support configurable evaluation criteria including:

- Price.
- Technical compliance.
- Supplier eligibility.
- Delivery capacity.
- Quality.
- Other approved criteria.

Conflict-of-interest declarations shall be supported.

FR-PRO-07 Award and Notification

The system shall support:

- Evaluation approval.
 - Award approval.
 - Supplier notification.
 - Unsuccessful supplier notification.
 - Grievance/clarification submission where configured.
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2.7 CONTRACT MANAGEMENT

FR-CON-01 Contract Creation

The system shall support:

- Contracts.
- Framework agreements.
- Purchase orders.
- Call-off orders.

FR-CON-02 Contract Balances

The system shall track:

- Contract value.
- Contract quantity.
- Used value.

- Used quantity.
- Remaining balance.

FR-CON-03 Contract Lifecycle

The system shall support:

- Drafting.
- Approval.
- Activation.
- Amendment.
- Suspension.
- Expiry.
- Closure.

FR-CON-04 Contract Monitoring

The system shall generate alerts for:

- Contract expiry.
- Remaining balance.
- Missed delivery milestones.
- Performance concerns.

2.8 AGGREGATION AND QUALITY ASSURANCE

FR-AGG-01 Produce Intake

Aggregators shall record:

- Supplier/source.
- Commodity.
- Quantity.
- Date.
- Location.
- Batch.

FR-AGG-02 Batch Management

Each batch shall have a unique identifier.

The architecture shall reserve the capability for future QR/barcode identification.

FR-AGG-03 Quality Inspection

Inspectors shall record:

- Inspection date.
- Inspector.
- Commodity.
- Quantity.
- Quality parameters.
- Results.
- Photographs.
- Acceptance/rejection.
- Corrective actions.

The system shall support offline capture where required.

FR-AGG-04 Traceability

The platform shall maintain traceability between:

Supplier/Producer → Aggregation Point → Batch → Dispatch → School Receipt

Advanced QR/barcode scanning may be implemented in a later phase. The system review identifies QR/barcode traceability as a later capability.

2.9 INVENTORY AND WAREHOUSING

FR-INV-01 Warehouse Management

The system shall maintain:

- Warehouse.
- Location.
- Capacity.
- Responsible officer.
- Status.

FR-INV-02 Goods Receipt

Warehouse users shall record:

- Delivery.

- Commodity.
- Quantity.
- Batch.
- Date.
- Supplier.
- Quality status.

FR-INV-03 Stock Movements

The system shall support:

- Receipts.
- Issues.
- Transfers.
- Adjustments.
- Returns.
- Rejections.
- Waste.

FR-INV-04 Stock Reservations

Stock may be reserved against:

- Purchase orders.
- Delivery plans.
- Schools.
- Other approved allocations.

FR-INV-05 Batch and Expiry

Where applicable, the system shall track:

- Batch number.
- Production date.
- Expiry date.
- Quantity remaining.

FR-INV-06 Stock Reconciliation

The system shall support:

- Physical counts.
- System balance.
- Variance.

- Adjustment approval.

FR-INV-07 Stock Alerts

Alerts shall support:

- Low stock.
- Expiring stock.
- Expired stock.
- Unusual stock movements.

Inventory and warehouse lifecycle management was identified as a major gap in the review.

2.10 LOGISTICS AND DELIVERY

FR-LOG-01 Dispatch Planning

Users shall create dispatch plans containing:

- Source.
- Destination.
- Commodity.
- Quantity.
- Batch.
- Vehicle.
- Driver.
- Delivery date.

FR-LOG-02 Route Management

The system shall support route planning.

GIS and route optimization shall be reserved for later implementation.

FR-LOG-03 Delivery Milestones

The system shall track:

- Dispatch.
- In transit.
- Arrived.
- Received.

- Rejected/returned.

FR-LOG-04 Proof of Delivery

Schools shall confirm:

- Quantity received.
- Quality status.
- Date/time.
- Receiver.
- Delivery documentation.
- Photographic evidence where required.

FR-LOG-05 Exceptions

Users shall record:

- Late delivery.
- Short delivery.
- Damaged goods.
- Rejected goods.
- Missing documents.
- Other exceptions.

FR-LOG-06 Replacement/Return

The system shall support approved replacement and return workflows.

2.11 FINANCE

FR-FIN-01 Invoice Submission

Invoices shall be linked to:

- Supplier.
- Purchase order/contract.
- Delivery.
- Accepted quantity.

FR-FIN-02 Three-Way Matching

The system shall compare:

Purchase Order ↔ Goods Receipt ↔ Invoice

Invoices with discrepancies shall be flagged.

FR-FIN-03 Invoice Approval

Invoices shall follow configurable approval workflows.

FR-FIN-04 Deductions and Adjustments

The system shall support:

- Partial acceptance.
- Deductions.
- Credit adjustments.
- Approved exceptions.

FR-FIN-05 Payment Status

The system shall track:

- Submitted.
- Approved.
- Processing.
- Partially Paid.
- Paid.
- Rejected.
- On Hold.

FR-FIN-06 Payment Integration

Payment status may initially be entered manually or imported through secure files.

API integration with financial/payment systems may be implemented later.

2.12 COMPLAINTS AND PERFORMANCE

FR-CMP-01 Complaint Management

Users shall create complaints containing:

- Complaint type.
- Description.
- Related transaction.

- Complainant.
- Date.
- Supporting documents.
- Status.

FR-CMP-02 Case Management

Cases shall support:

- Assignment.
- Investigation.
- Action.
- Escalation.
- Resolution.
- Closure.

FR-CMP-03 Performance Management

Performance indicators shall be generated for:

- Suppliers.
 - Schools.
 - Counties.
 - Procurement teams.
 - Aggregators.
 - Logistics providers.
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2.13 MEAL AND ANALYTICS

FR-MEAL-01 Dashboards

Dashboards shall be available according to user role and access scope.

FR-MEAL-02 Programme Indicators

The platform shall support indicators relating to:

- Procurement.
- Supplier participation.
- Inclusive procurement.
- Delivery.
- Quality.

- Payment.
- School coverage.
- Commodity availability.

FR-MEAL-03 Reporting and Export

Authorized users shall export data in standard formats such as:

- CSV.
- Excel.
- PDF where required.

Advanced forecasting, price intelligence and anomaly detection shall be introduced in later phases.

2.14 PLATFORM SERVICES

FR-PLT-01 Workflow Engine

The platform shall provide reusable workflows supporting:

- Approval stages.
- Threshold-based routing.
- Delegation.
- Acting officers.
- Escalation.
- SLA monitoring.
- Rejection.
- Resubmission.
- Approval history.

FR-PLT-02 Notification Service

The platform shall support:

- SMS.
- Email.
- In-app notifications.
- Push notifications where supported.

Notifications shall be event-driven.

Examples:

- New sourcing opportunity.
- Approval required.
- Document expiry.
- Contract expiry.
- Delivery exception.
- Payment status change.

FR-PLT-03 Document Management

Documents shall support:

- Document type.
- Version.
- Transaction linkage.
- Expiry date.
- Access permissions.
- Secure storage.
- Retention.
- Audit trail.

FR-PLT-04 Audit Trail

All material transactions shall be auditable.

FR-PLT-05 System Configuration

Authorized administrators shall manage configurable:

- Thresholds.
- Workflows.
- Notifications.
- Reference data.
- SLAs.
- Business rules.

2.15 RISK AND EXCEPTION MANAGEMENT

This capability is included as a cross-cutting function rather than a separate primary business domain.

FR-RSK-01 Risk Identification

The system shall identify configurable risk conditions such as:

- Supplier eligibility concerns.
- Repeated late deliveries.
- Quality failures.
- Procurement exceptions.
- Budget overruns.
- Payment delays.

FR-RSK-02 Risk Flags

Transactions and suppliers may receive risk flags.

FR-RSK-03 Exception Management

Exceptions shall support:

- Identification.
- Assignment.
- Investigation.
- Approval.
- Corrective action.
- Closure.

FR-RSK-04 Corrective Actions

Users shall create and monitor corrective actions.

FR-RSK-05 Risk Escalation

High-priority exceptions shall be escalated according to configured rules.

FR-RSK-06 Risk Dashboard

Authorized users shall view risk and exception summaries.

The review identified centralized risk and exception management as a gap in the original brief.

3. NON-FUNCTIONAL REQUIREMENTS

NFR-01 Availability

Target availability:

99.5% uptime

Final SLA to be confirmed.

NFR-02 Performance

The system should target:

- Page/API response within approximately 3 seconds for 95% of normal requests under agreed load.
- Fast response on low-bandwidth connections.
- Asynchronous processing for heavy operations.

The final performance target shall be validated during performance testing.

NFR-03 Scalability

The platform shall support:

- Multiple counties.
- Multiple schools.
- Multiple suppliers.
- Multiple procurement events.
- Increasing transaction volumes.

NFR-04 Security

The system shall implement:

- TLS.
- Encryption at rest.
- Secure authentication.
- RBAC.
- MFA for privileged users.
- Secure password storage.
- Input validation.
- Protection against common web attacks.
- Security logging.

NFR-05 Accessibility

The system shall target:

WCAG 2.1 AA

The interface shall support:

- Clear labels.
- Keyboard navigation where applicable.
- Sufficient contrast.
- Accessible forms.
- Screen-reader-friendly structure.

NFR-06 Mobile and Low-Bandwidth Support

The system shall work on:

- Standard desktop browsers.
- Mobile browsers.
- Low-end Android devices.
- Low-bandwidth networks.

NFR-07 Offline Capability

Offline capability shall support defined workflows for selected users.

MVP offline functionality shall focus on:

- Offline data capture.
- Local queueing.
- Sync when connectivity returns.
- Retry handling.
- Sync status.

Advanced conflict-resolution mechanisms shall be introduced in later phases.

This distinction resolves the previous ambiguity between “basic offline capture” in the MVP and “full offline synchronization with conflict resolution” as a later feature.

NFR-08 Auditability

Transactions shall be traceable from creation to closure.

NFR-09 Data Retention

Retention periods shall be configurable and confirmed during inception.

4. ROLES AND PERMISSIONS

Role	Main Responsibility
Super Administrator	Overall system administration
System Administrator	Technical configuration and users
County Administrator	County-level oversight
County Procurement Officer	County procurement
County Finance Officer	County budget and finance
School Administrator	School administration
School Meals Officer	Meal planning and demand
Procurement Officer	Procurement operations
Evaluation Committee Member	Bid evaluation
Quality Inspector	Quality inspection
Aggregator	Produce aggregation and batch management
Supplier	Registration, bids and fulfilment
Warehouse Officer	Inventory
Logistics Officer/Driver	Dispatch and delivery
Finance Officer	Invoice/payment processing
Programme/MEAL Officer	Monitoring and reporting
Auditor	Audit and review

Additional roles may be configured during inception where required, including:

- Nutrition Officer.
- Approving Officer.
- Risk/Compliance Officer.
- Helpdesk Officer.

These additional roles should only be activated if the operating model requires them.

4.1 Permission Model

Permissions shall be structured as:

User → Role → Organization/Data Scope → Permission

Example:

John → County Procurement Officer → County A → Create/Edit/Submit
Procurement

This prevents a user from receiving unrestricted access simply because they have a particular role.

5. WORKFLOW AND APPROVAL ENGINE

5.1 End-to-End Workflow

The recommended transaction flow is:

School Demand

↓

Demand Calculation

↓

Demand Consolidation

↓

Budget Validation

↓

Procurement Plan

↓

Approval

↓

RFQ/Sourcing Event



Supplier Bid



Bid Closing



Evaluation



Approval/Award



PO/Contract



Aggregation



Quality Inspection



Inventory



Dispatch



Delivery



School Receipt



Three-Way Match



Invoice Approval



Payment



Supplier/School Performance



MEAL Reporting

This end-to-end workflow reflects the recommended system workflow in the review.

5.2 Transaction State Model

Every major transaction shall have a defined lifecycle.

Demand

Draft → Submitted → Validated → Approved → Converted to Procurement → Closed

Procurement Event

Draft → Published → Open → Closed → Under Evaluation → Approved → Awarded → Closed

Bid

Draft → Submitted → Locked → Under Evaluation → Accepted/Rejected

Purchase Order

Draft → Submitted → Approved → Issued → Partially Fulfilled → Fulfilled/Closed → Cancelled

Contract

Draft → Under Review → Approved → Active → Suspended/Amended → Expired/Closed

Batch

Created → Received → Inspected → Accepted/Rejected → Allocated →
Dispatched → Delivered

Delivery

Planned → Dispatched → In Transit → Delivered → Accepted/Rejected →
Closed

Invoice

Draft → Submitted → Matched → Approved → Processing → Paid/Partially
Paid/Rejected

Each state transition shall be:

- Authorized.
- Logged.
- Timestamped.
- Linked to the responsible user.
- Validated against business rules.

5.3 Business Rules Catalogue

Business rules shall be maintained separately from functional requirements.

Examples:

BR-001 Budget Availability

A procurement transaction shall not proceed beyond the configured commitment stage where sufficient approved budget is unavailable, unless an approved exception exists.

BR-002 Supplier Eligibility

Only suppliers meeting the configured eligibility requirements may participate in restricted sourcing events.

BR-003 Bid Closing

A bid cannot be modified after the sourcing event closing time.

BR-004 Approval Segregation

A user cannot approve a transaction where the configured segregation-of-duties rule prohibits self-approval.

BR-005 Delivery

An invoice cannot proceed to normal approval unless the associated goods receipt has been recorded and accepted, subject to approved exceptions.

BR-006 Three-Way Match

The system shall compare purchase order, receipt and invoice quantities/values before invoice approval.

BR-007 Stock

Stock balances shall not become negative unless negative stock is explicitly permitted by configuration.

BR-008 Contract Balance

A purchase order/call-off shall not exceed the available contract quantity/value unless an approved variation exists.

BR-009 Document Expiry

Expired mandatory supplier documents shall trigger configurable restrictions or alerts.

BR-010 Audit

Material transaction changes shall generate an audit record.

The final business-rule catalogue shall be validated with procurement, finance and programme stakeholders before development.

5.4 Acceptance Criteria

Major requirements shall have testable acceptance criteria.

Example:

Procurement Event

The feature is accepted when:

1. Authorized users can create a sourcing event.
 2. Eligible suppliers can be notified.
 3. Suppliers can submit bids before the deadline.
 4. The system automatically closes the event.
 5. Late bids are blocked.
 6. Authorized evaluators can access submitted bids.
 7. Evaluation results are recorded.
 8. Approval follows configured thresholds.
 9. The award is recorded.
 10. All actions are auditable.
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6. DATA MODEL

The database shall use PostgreSQL as the primary relational database.

PostGIS shall be available if GIS functionality is adopted.

6.1 Core Entities

Identity

- User
- Role
- Permission
- Organization
- UserRole
- Delegation
- Session

Master Data

- County
- SubCounty
- Ward
- School
- SchoolCluster
- Commodity
- Unit
- Menu

- NutritionSpecification
- QualityStandard

Demand

- SchoolDemand
- DemandLine
- MenuPlan
- StockBalance
- DemandConsolidation
- ProcurementPlan

Budget

- FundingSource
- Budget
- BudgetLine
- Commitment
- BudgetTransaction
- BudgetException

Supplier

- Supplier
- SupplierContact
- SupplierDocument
- SupplierCertification
- SupplierPrequalification
- SupplierCapacity
- SupplierPerformance

Procurement

- ProcurementEvent
- ProcurementLot
- Bid
- BidLine
- Evaluation
- EvaluationCriterion
- Award
- PurchaseOrder

Contract

- Contract
- ContractLine
- ContractAmendment
- CallOff

Aggregation

- AggregationCentre
- Intake
- Batch
- QualityInspection
- QualityResult
- CorrectiveAction

Inventory

- Warehouse
- StockItem
- StockMovement
- StockReservation
- StockTransfer
- StockAdjustment
- StockCount

Logistics

- Dispatch
- DispatchLine
- Route
- Vehicle
- Driver
- Delivery
- ProofOfDelivery
- DeliveryException

Finance

- Invoice
- InvoiceLine
- MatchResult
- Payment
- PaymentStatus
- Deduction

Complaints

- Complaint
- Case
- CaseAction
- Escalation

Platform

- Notification
 - Document
 - AuditLog
 - Workflow
 - WorkflowInstance
 - Approval
 - SystemConfiguration
 - IntegrationTransaction
 - SyncTransaction
 - RiskFlag
 - Exception
-

6.2 Data Constraints

The database shall enforce appropriate:

- Primary keys.
- Foreign keys.
- Unique constraints.
- Not-null constraints.
- Check constraints.
- Status constraints.
- Referential integrity.

Examples:

- Quantity must be greater than zero.
- Dates must follow logical sequence.
- Supplier identifiers must be unique.
- Batch identifiers must be unique within the required scope.
- Invoice lines must reference valid transactions.
- Contract call-offs cannot exceed approved balances.

- School records must belong to a valid geographic/organizational structure.
-

6.3 Common Data Standards

The platform shall use consistent standards for:

- UUIDs/identifiers.
- Dates and timestamps.
- Currency.
- Units of measure.
- Phone numbers.
- Country codes.
- Status values.
- Geographic codes.

All timestamps shall be stored consistently, preferably in UTC, with user-facing conversion to the relevant local timezone.

6.4 Search, Filtering and Pagination

All major list screens shall support, where applicable:

- Search.
- Filtering.
- Sorting.
- Pagination.
- Date filtering.
- Status filtering.
- Export.

This should be implemented consistently across the platform.

7. SECURITY, PRIVACY AND COMPLIANCE

7.1 Authentication

The system shall support secure authentication using OAuth2/OIDC-compatible mechanisms.

7.2 Authorization

Authorization shall be enforced at the API level, not only at the user-interface level.

7.3 Encryption

The platform shall use:

- TLS for data in transit.
- Encryption for sensitive data at rest.
- Secure password hashing.

7.4 Audit Logging

Audit records should contain, where applicable:

- User ID.
- User role.
- Organization.
- Timestamp.
- Action.
- Entity.
- Entity ID.
- Previous state.
- New state.
- Reason.
- Source/device information where appropriate.
- Correlation ID.

7.5 Privacy

The platform shall align with applicable Kenyan data protection requirements, including the Data Protection Act, 2019.

7.6 Security Testing

Before production release, the system shall undergo:

- Vulnerability scanning.
- Dependency scanning.
- Authentication/authorization testing.
- API security testing.
- Penetration testing where appropriate.
- Remediation and retesting.

8. INTEGRATION DESIGN

8.1 Integration Targets

Potential integrations include:

- SMS provider.
- Email provider.
- Mobile money/payment systems.
- Banking systems.
- County financial systems.
- Government procurement systems.
- School information systems.

The exact systems will be confirmed during inception.

8.2 API Architecture

The platform shall expose versioned REST APIs.

Example:

```
/api/v1/users
```

```
/api/v1/schools
```

```
/api/v1/suppliers
```

```
/api/v1/procurement-events
```

```
/api/v1/bids
```

```
/api/v1/purchase-orders
```

```
/api/v1/deliveries
```

```
/api/v1/invoices
```

APIs shall have:

- Authentication.
- Authorization.
- Validation.

- Standard response structures.
 - Standard error structures.
 - Pagination.
 - Filtering.
 - Sorting.
 - Rate limiting where required.
 - Idempotency for appropriate transaction endpoints.
 - Correlation IDs.
 - OpenAPI documentation.
-

8.3 Error Response Standard

The API shall not expose raw database or infrastructure errors to users.

Example response structure:

```
{
  "success": false,
  "error": {
    "code": "BUDGET_EXCEEDED",
    "message": "The available budget is insufficient for this transaction."
    "details": []
  },
  "correlation_id": "...
}
```

8.4 Idempotency

Idempotency shall be implemented for operations where repeated requests could create duplicate transactions.

Examples:

- Payment callbacks.
 - Invoice submission.
 - Purchase-order creation.
 - Offline synchronization.
 - External integration callbacks.
-

8.5 File and Document Management

Uploaded documents shall support:

- Approved file types.
- Maximum file size.
- Malware scanning.
- File checksum.
- Versioning.
- Secure storage.
- Access control.
- Expiry dates.
- Transaction linkage.
- Retention rules.

Files shall be stored in S3-compatible object storage rather than directly inside the relational database where appropriate.

9. TECHNICAL ARCHITECTURE

9.1 Frontend

Recommended:

- React.
- Next.js.
- TypeScript.
- Responsive design.
- Progressive Web App capability.

9.2 Backend

Recommended:

- Python.
- FastAPI.
- Pydantic.
- SQLAlchemy.

9.3 Database

Primary:

PostgreSQL

GIS-ready:

PostGIS

9.4 File Storage

S3-compatible object storage.

9.5 Background Processing

Redis plus a task/queue service shall support:

- Notifications.
- Scheduled jobs.
- Reports.
- Alerts.
- Synchronization.
- Other asynchronous operations.

9.6 Deployment

The platform shall be containerized.

Recommended environments:

1. Development
2. QA/Test
3. Staging/UAT
4. Production

Production credentials, databases and secrets shall be separated from development environments.

Production data shall not be copied into development environments unless appropriately anonymized.

9.7 CI/CD

The development pipeline shall support:

- Source control.
- Automated tests.

- Code quality checks.
 - Security scanning.
 - Build automation.
 - Deployment automation.
 - Database migration management.
 - Rollback procedures.
-

9.8 Observability

The platform shall provide centralized:

- Application logs.
- Infrastructure metrics.
- Error monitoring.
- API performance monitoring.
- Database monitoring.
- Queue monitoring.
- Notification monitoring.
- Offline synchronization monitoring.

Alerts should be generated for critical failures.

10. OFFLINE SYNCHRONIZATION DESIGN

10.1 MVP Offline Capability

The MVP shall support selected offline workflows.

Offline records shall contain:

- Local transaction ID.
- Server transaction ID where available.
- User ID.
- Device/session ID where appropriate.
- Created timestamp.
- Updated timestamp.
- Version.
- Sync status.

10.2 Sync States

Example:

Pending → Uploading → Synced

or

Pending → Failed → Retry → Synced

10.3 Retry

Failed synchronization shall support automatic or manual retry.

10.4 Conflict Handling

The MVP shall use clearly defined conflict rules.

Advanced conflict detection and resolution shall be introduced in a later phase.

10.5 Offline Attachments

Where required, users shall be able to capture:

- Photographs.
- Inspection evidence.
- Proof of delivery.

Files shall be queued for upload when connectivity returns.

The system review specifically identifies offline storage, queued transactions, automatic sync, conflict detection, conflict resolution and offline photo capture as capabilities requiring explicit design.

11. MVP AND DEVELOPMENT PHASES

Phase 0 – Requirements and Design

Deliver:

- Validated requirements.
- Process maps.
- Business rules.
- ERD.
- Database specification.
- UI prototype.

- Security architecture.
- Integration specification.
- Pilot configuration.

Phase 1 – Platform Foundation

Develop:

- Authentication.
- RBAC.
- Organizations.
- Schools.
- Master data.
- Suppliers.
- Documents.
- Audit trail.
- Configuration.

Phase 2 – Demand and Budget

Develop:

- School demand.
- Menus.
- Stock.
- Demand calculations.
- Consolidation.
- Procurement plans.
- Budget controls.

Phase 3 – e-Procurement

Develop:

- RFQs.
- Sourcing events.
- Supplier notification.
- Bid submission.
- Evaluation.
- Approvals.
- Awards.
- Purchase orders.
- Contract management.

Phase 4 – Supply Chain Fulfilment

Develop:

- Aggregation.
- Batches.
- Quality.
- Inventory.
- Dispatch.
- Logistics.
- Delivery.
- Proof of delivery.

Phase 5 – Finance and Performance

Develop:

- Invoice processing.
- Three-way matching.
- Payment tracking.
- Reconciliation.
- Supplier performance.
- Complaints.
- MEAL dashboards.

Phase 6 – Advanced and Scale

Develop:

- Advanced offline synchronization.
- GIS.
- QR/barcode traceability.
- Route optimization.
- Advanced inventory.
- Forecasting.
- Price intelligence.
- Risk/anomaly detection.
- Deeper integrations.
- Data warehouse/BI.
- AI-assisted decision support.

This phased approach follows the recommended development sequence in the system review.

11.1 MVP Scope

The MVP shall include:

- Authentication.
- Role-based access.
- Master data.
- School management.
- Supplier registration.
- Supplier prequalification.
- Demand planning.
- Procurement planning.
- Budget validation.
- RFQs/sourcing.
- Bid submission.
- Bid evaluation.
- Approval workflows.
- Purchase orders.
- Basic aggregation.
- Quality inspection.
- Delivery confirmation.
- Invoice tracking.
- Three-way matching.
- Payment status.
- Notifications.
- Audit trail.
- Core dashboards.
- Data exports.
- Basic offline capture for selected workflows.

The system review recommends keeping advanced capabilities out of the first MVP to avoid excessive implementation complexity.

11.2 Testing Strategy

Testing shall include:

Unit Testing

Test individual functions and services.

Integration Testing

Test communication between:

- Frontend/backend.
- Backend/database.
- Notification services.
- External integrations.

System Testing

Test complete workflows.

End-to-End Testing

Example:

Demand → Budget → RFQ → Bid → Evaluation → PO → Delivery → Invoice → Payment

Security Testing

Test:

- Authentication.
- Authorization.
- Session management.
- API security.
- File security.

Performance Testing

Test:

- Concurrent users.
- API performance.
- Database performance.
- Large exports.
- Notification queues.

Offline Testing

Test:

- Offline capture.
- Network loss.
- Sync.
- Retry.
- Duplicate prevention.

UAT

Representatives from:

- Schools.
- Procurement.
- Finance.
- Suppliers.
- Aggregators.
- Quality teams.
- Programme/MEAL.

shall participate in user acceptance testing.

11.3 Minimum Acceptance Criteria

The pilot system shall be considered functional when:

1. Authorized users can complete the sourcing cycle from demand entry to payment tracking.
2. Multiple schools can be managed.
3. Multiple counties can be supported.
4. Multiple commodities can be configured.
5. Multiple supplier types can be supported.
6. Supplier eligibility can be digitally verified.
7. Supporting documents can be uploaded and verified.
8. Procurement approvals follow configured workflows.
9. Transactions have complete audit trails.
10. Deliveries can be traced from school to supplier/aggregation point.
11. Payment status is visible to authorized users.
12. Dashboards provide accurate information.
13. Data can be exported.
14. The platform works on standard desktop and mobile devices.
15. User testing confirms usability.
16. Security testing is completed.

17. Backup and recovery testing is completed.

These criteria are consistent with the original system development brief's acceptance expectations.

11.4 Operational Support

The implementation shall define:

- System owner.
 - Technical owner.
 - Helpdesk.
 - Support levels.
 - Incident severity levels.
 - Escalation procedures.
 - Maintenance windows.
 - Backup responsibilities.
 - Security incident procedures.
 - User administration responsibilities.
-

11.5 Backup and Disaster Recovery

The system shall define:

- Backup frequency.
- Backup retention.
- Off-site backup.
- Database backup.
- File-storage backup.
- Restore procedures.
- Restore testing.
- Disaster recovery procedure.
- Business continuity procedure.

Final:

- RTO.
- RPO.
- Backup retention period.

shall be confirmed during inception.

11.6 Implementation Deliverables

The development team shall provide:

1. Business Requirements Document.
 2. System Requirements Specification.
 3. Process/workflow diagrams.
 4. UI/UX prototype.
 5. ERD.
 6. Database schema.
 7. API specification.
 8. Source code.
 9. Automated tests.
 10. Security assessment/report.
 11. Deployment configuration.
 12. CI/CD configuration.
 13. User manuals.
 14. Administrator manual.
 15. API documentation.
 16. Training materials.
 17. UAT report.
 18. Backup and DR documentation.
 19. Operational runbook.
 20. System handover documentation.
-

12. OPEN INCEPTION QUESTIONS

The following must be resolved before final development sign-off:

1. Which counties, schools and commodities are included in the pilot?
2. Which procurement methods must be supported?
3. Which financial/procurement thresholds must be enforced?
4. Who is the legal/institutional owner after the pilot?
5. Who controls master data?
6. Who controls approval workflows?
7. Which government, finance, banking, mobile-money and school systems must be integrated?

8. Which offline functions are mandatory?
9. Which users require PWA/mobile access?
10. Which nutrition standards and commodity specifications are authoritative?
11. Who performs and approves quality inspections?
12. What data must be retained and for how long?
13. Who can access sensitive information?
14. What hosting arrangement will be used?
15. What backup and disaster-recovery arrangements will apply?
16. What are the RTO and RPO?
17. What uptime SLA is required?
18. What are the support SLAs and escalation procedures?
19. Which languages are required?
20. Which additional operational roles are required?
21. Which payment status integration method will be used?
22. What financial definition will be used for commitment, obligation, expenditure and payment?

The original review identifies the pilot geography, procurement thresholds, system ownership, integrations, offline requirements, nutrition standards, quality responsibility, retention, hosting/DR and service levels as critical decisions before coding.

13. TRACEABILITY AND REQUIREMENTS CONTROL

Every functional requirement shall be traceable to one or more of:

- Original System Development Brief.
- Approved business process.
- Approved business rule.
- User requirement.
- Legal/compliance requirement.
- Technical requirement.
- System review recommendation.

Each requirement shall have:

- Requirement ID.
- Description.
- Source.
- Priority.

- Business owner.
- Development status.
- Test case.
- Acceptance criteria.

A requirements traceability matrix shall be maintained throughout development.

14. REQUIREMENT PRIORITY

Requirements shall be classified as:

MUST HAVE

Required for MVP and pilot operation.

SHOULD HAVE

Important for operational effectiveness but may follow immediately after MVP.

COULD HAVE

Useful enhancements that do not block the core system.

FUTURE

Advanced capabilities intended for later phases.

This prioritization shall be reviewed during inception and may change based on pilot needs, budget and implementation capacity.

15. RECOMMENDED SYSTEM PRINCIPLE

LisheBora should not be developed as only an electronic tendering platform.

The target architecture is an integrated school food supply-chain platform with e-procurement at its core:

**Demand → Budget → Sourcing → Evaluation → Contract → Aggregation →
Quality → Inventory → Logistics → Delivery → Invoice → Payment →
Performance → MEAL**

The system should be modular, configurable, API-enabled, secure, mobile-friendly and capable of operating in low-connectivity environments.

Advanced capabilities should be added progressively without requiring a redesign of the core platform.

16. FINAL DESIGN POSITION

The LisheBora e-Sourcing Platform specification is now structured as a developer-oriented system specification rather than only a business requirements document.

The development team should not proceed directly to full coding from this document alone. Before implementation, the project should complete the inception/design stage and produce the following approved artefacts:

Requirements → Process Maps → Business Rules → ERD → UI Prototype → API Specification → Security Design → Integration Design → Test Plan → Deployment Plan

Once these are approved, development can proceed in phases while maintaining traceability between requirements, code, testing and user acceptance.

This **v0.2 is substantially more developer-ready** than the previous version. I have also kept advanced features such as **GIS, QR/barcode traceability, AI, advanced forecasting and full offline conflict resolution** out of the MVP, in line with the system review rather than making the first version unnecessarily large.

LisheBora_System_Document_Revie...

The next logical step would be to turn this into a **developer handover package** with the **ERD/database tables,** **API endpoint catalogue,** **detailed user stories,** **acceptance criteria/test cases, and** **screen-by-screen UI structure.**

Sources